

PM2.5 & Asthma Analysis

Pragya, Shradha, Sania, Anish

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Overview

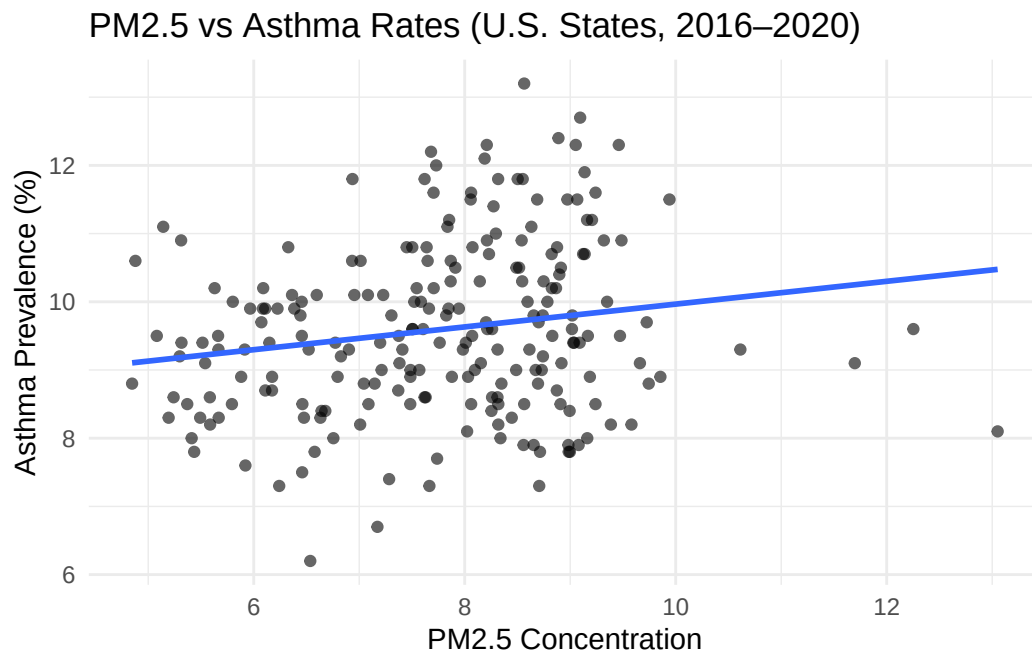
This report analyzes the relationship between PM2.5 air pollution levels and asthma prevalence across U.S. states from 2016–2020. The goal is to explore whether higher air pollution is associated with higher asthma rates.

Load and Merge Data

Preview Data

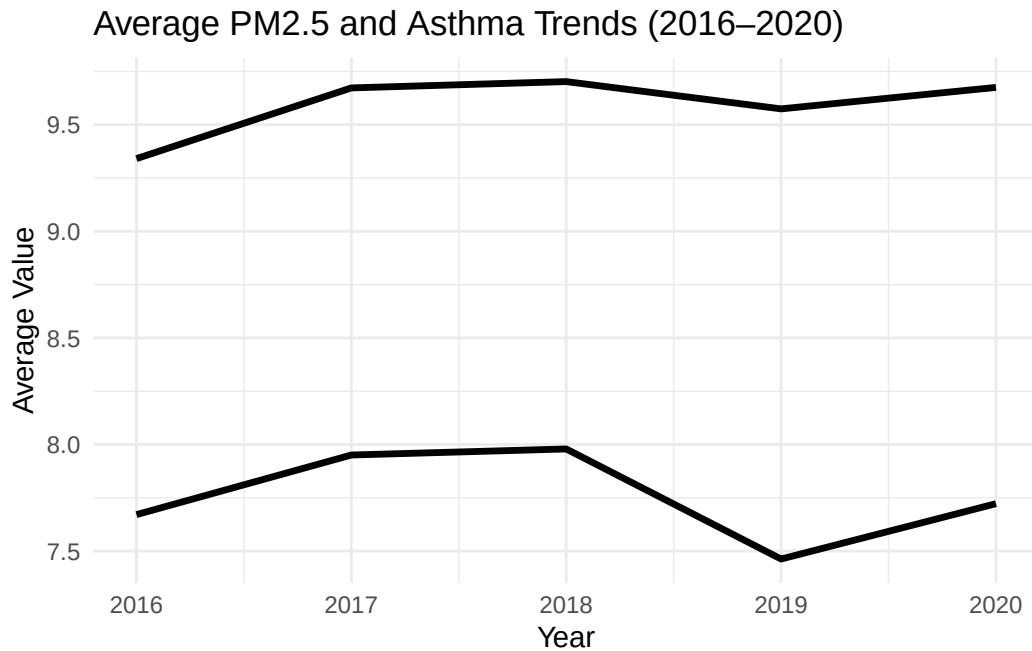
```
# A tibble: 6 x 5
  statefips year pm25 state asthma_rate
  <dbl> <dbl> <dbl> <chr> <dbl>
1         1  2016  9.73 AL          9.7
2         1  2017  9.32 AL         10.9
3         1  2018  8.91 AL         10.5
4         1  2019  9.17 AL          9.5
5         1  2020  8.73 AL           9
6         4  2016  7.09 AR          8.5
```

PM2.5 vs Asthma Relationship



Trends Over Time

This plot shows how average PM2.5 levels and asthma rates changed over time.



Correlation

[1] 0.1803415

Key Findings

The scatterplot suggests a positive relationship between PM2.5 levels and asthma rates, meaning that states with higher air pollution tend to have higher asthma prevalence. However, the relationship is not perfectly linear, indicating that other factors may also influence asthma outcomes.

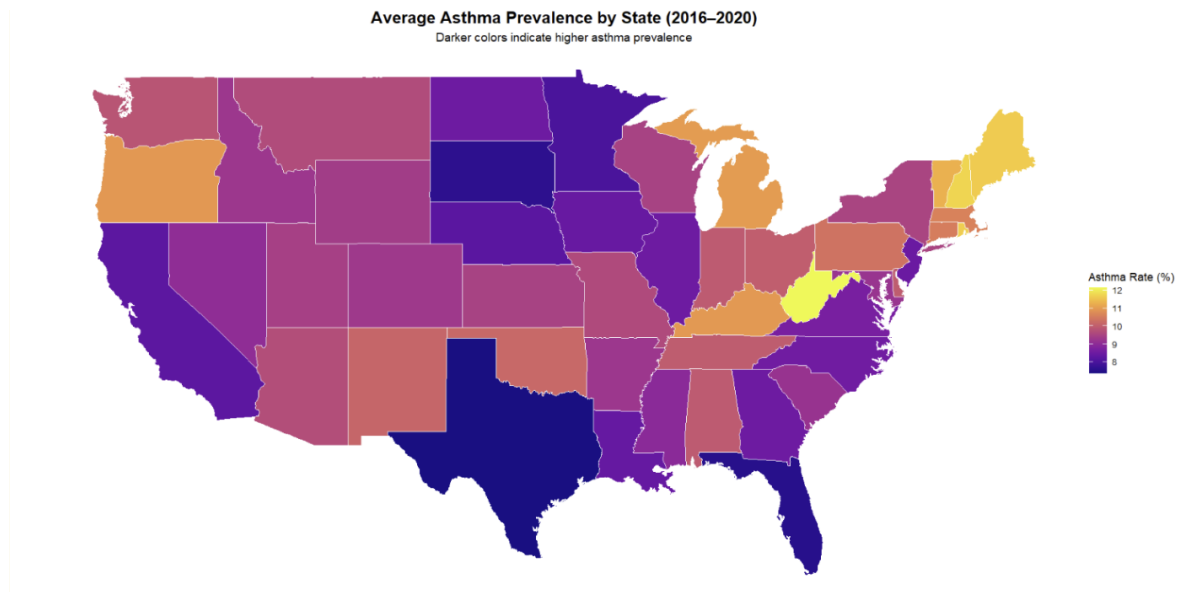
The time trend plot shows that both PM2.5 levels and asthma rates fluctuate over time, with no dramatic drop in 2020 despite reduced activity during the COVID-19 pandemic. This suggests that air pollution and asthma are influenced by multiple long-term factors beyond short-term changes.

Summary

This analysis provides evidence of a potential association between air pollution and asthma prevalence across U.S. states. While higher PM2.5 levels generally correspond to higher asthma rates, additional factors such as healthcare access, demographics, and environmental policies likely play a role. Further analysis could explore these relationships in more depth.

Geographic Distribution of PM2.5 and Asthma

This section visualizes how PM2.5 pollution and asthma prevalence vary across U.S. states.



This map shows that certain states consistently experience higher levels of air pollution and asthma prevalence. Regions with elevated PM2.5 levels often overlap with areas that report higher asthma prevalence, suggesting a possible geographic pattern in environmental health risks. These patterns highlight potential environmental health disparities across regions. Further analysis could examine regional policy impacts.